Here is another "orthogonality formula" for characters, in which summation is taken over irreducible representations rather than group elements.

Theorem 4.5.4. *Let $g, h \in G$, and let Z_g denote the centralizer of g in G. Then*

$$\sum_V \chi_V(g)\overline{\chi_V(h)} = \begin{cases} |Z_g|, & \text{if } g \text{ is conjugate to } h, \\ 0, & \text{otherwise,} \end{cases}$$

where the summation is taken over all irreducible representations of G.

Proof. As noted above, $\overline{\chi_V(h)} = \chi_{V^*}(h)$, so the left-hand side equals (using Maschke's theorem):

$$\begin{aligned}\sum_V \chi_V(g)\chi_{V^*}(h) &= \operatorname{Tr}|_{\bigoplus_V V\otimes V^*}(g \otimes (h^*)^{-1}) \\ &= \operatorname{Tr}|_{\bigoplus_V \operatorname{End} V}(x \mapsto gxh^{-1}) = \operatorname{Tr}|_{\mathbb{C}[G]}(x \mapsto gxh^{-1}).\end{aligned}$$

If g and h are not conjugate, this trace is clearly zero, since the matrix of the operator $x \mapsto gxh^{-1}$ in the basis of group elements has zero diagonal entries. On the other hand, if g and h are in the same conjugacy class, the trace is equal to the number of elements x such that $x = gxh^{-1}$, i.e., the order of the centralizer Z_g of g. We are done. □

Remark 4.5.5. Another proof of this result is as follows. Consider the matrix U whose rows are labeled by irreducible representations of G and whose columns are labeled by conjugacy classes, with entries $U_{V,g} = \chi_V(g)/\sqrt{|Z_g|}$. Note that the conjugacy class of g is G/Z_g; thus $|G|/|Z_g|$ is the number of elements conjugate to g. Thus, by Theorem 4.5.1, the rows of the matrix U are orthonormal. This means that U is unitary and hence its columns are also orthonormal, which implies the statement.